The exact-count falsifier for lever 2: no candidate family fires it, and none differs from random bases

2026-11-01. Protocol frozen at `9d2a2daf2`, before any row (`PROTOCOL.md`). The count asked for by `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §4 item 2, run exactly: for a base $X \subset F_p$ of size B, the closure count $c_m(r)$ of $S_{m+1}(x_1, \ldots, x_m, r) = 0$ over every ordered tuple of X and every target $r \in F_p$, compared with the threshold B^{m-1}.

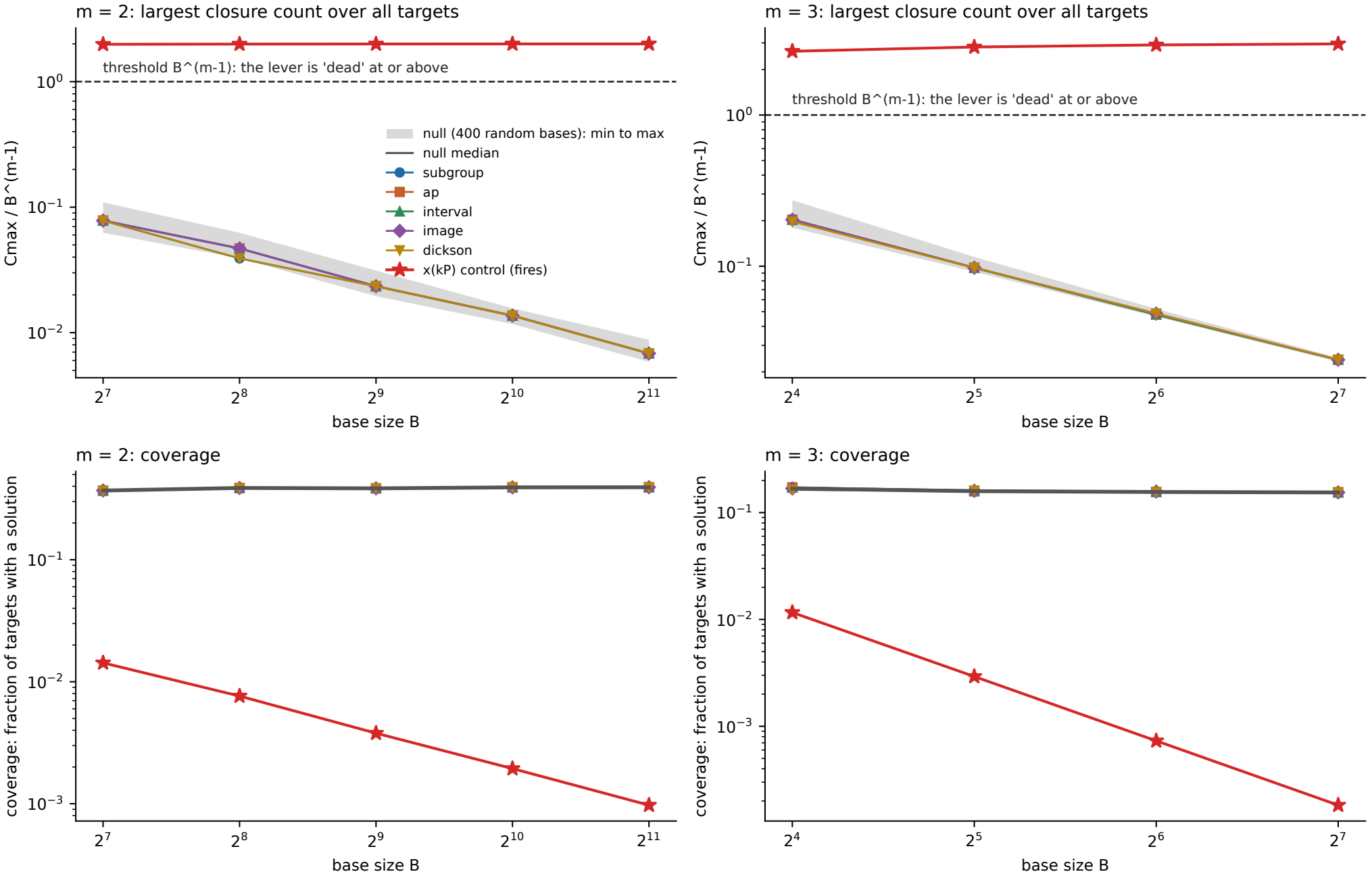
Result

- **Instrument checked.** The fast count equals brute-force evaluation at every r for $m = 2, 3$ and all seven base families (G1). S_3 vanishes at $x(P_1 + P_2)$ over 9,900 point pairs and S_4 at $x(P_1 + P_2 + P_3)$ over 83,285 point triples (G2). S_4 is symmetric and its polynomial form equals its numeric value (G3). Every base element carries the $3B - 2$ trivial tuples (x, x, r) at $m = 3$ in all 2,560 $m = 3$ instances (G5, `trivial_violations = 0`). Clippy with `-D warnings` and rustfmt are clean.
- **Exact, no sampling.** 5,760 instances: 3,200 at $m = 2$ over $B = 128 \ldots 2048$ and 2,560 at $m = 3$ over $B = 16 \ldots 128$ (p up to 4,206,593), each counted over all p targets.
- **The literal falsifier does not fire for any candidate family.** None of the 1,800 structured instances (subgroup coset, arithmetic progression, interval, cubic image, Dickson image; 40 per family per size) reaches B^{m-1}. The median of their largest counts is 0.7%–8% of the threshold at $m = 2$ and 2.4%–20% at $m = 3$, and that share falls as B grows. The largest single instance reaches 14 of 128 at $m = 2$ and 64 of 256 at $m = 3$.
- **None differs from random bases.** All 90 Mann–Whitney tests of a family against the 400-instance null, on C_{max} and coverage, are within the null at $\alpha_c = 0.01/90 \approx 1.1 \cdot 10^{-4}$. The smallest family p-value is $4.3 \cdot 10^{-3}$ (image, coverage, $m = 2$, $B = 2048$) and the largest $|\text{Cliff's } \delta|$ is 0.27.
- **The detector sees concentration.** The control $x(kP)$ fires in 40 of 40 instances at each of the nine sizes (360 of 360), and its coverage is below the null's at $p \approx 10^{-25}$. Its C_{max} is exactly $2B - 2$ at $m = 2$ and exactly $3B^2 - 6B + 4$ at $m = 3$ at all nine sizes (fit to the recorded values, not derived here), while its coverage falls to 0.1% and 0.02% of targets.
- **The A/A split of the null passes** at every size (18 tests, smallest $p = 0.12$).

1. Claim classes

claim	class	basis
each $c_m(r)$, C_{max}, coverage and firing flag of each instance	exact computation	counted over all tuples and targets; gates G1–G5
no structured instance has $C_{max} \geq B^{m-1}$; the control always does	exact computation	`fires` per instance in `rows.jsonl`
a family does or does not differ from the null	comparative, statistical	controls of §5 below; 90 tests
the closed forms of the control's C_{max}	observation	fit to nine recorded values
growth of C_{max} / B^{m-1} with p	descriptive	the figure, no test
lever viability, any attack cost	out of scope	not established

2. Figure



Exact counts over every ordered tuple of the base and every target in F_p (runs/m*/summary.json); 40 instances per family, 400 for the null, p the smallest prime 2Bk+1 >= B^m.

Figure 1: Largest closure count over the threshold (top) and coverage (bottom), by base size; m = 2 left, m = 3 right. Exact counts from the summary.json of runs/m2 and runs/m3.

Top: the largest closure count over all targets, divided by the threshold B^{m−1}; the dashed line is the threshold. Grey: min to max over the 400 random bases. The five structured families sit on the null's median; the control sits above the threshold. Bottom: the fraction of targets with at least one solution.

3. Summary by size

m	B	p	threshold	null Cmax: median [min, max]	structured Cmax: range of family medians	structured fires	smallest family p	control Cmax (fires)
2	128	17921	128	10 [8, 14]	10 to 10	0/200	5.8e-02	254 (40/40)
2	256	67073	256	12 [10, 16]	10 to 12	0/200	3.8e-02	510 (40/40)
2	512	270337	512	12 [10, 16]	12 to 12	0/200	5.4e-02	1022 (40/40)
2	1024	1054721	1024	14 [12, 16]	14 to 14	0/200	2.1e-02	2046 (40/40)
2	2048	4206593	2048	14 [12, 18]	14 to 14	0/200	4.3e-03	4094 (40/40)
3	16	4129	256	52 [46, 70]	50.5 to 52	0/200	5.9e-02	676 (40/40)
3	32	32833	1024	100 [94, 118]	100 to 100	0/200	4.7e-02	2884 (40/40)
3	64	262657	4096	196 [190, 214]	196 to 199	0/200	1.1e-01	11908 (40/40)
3	128	2100737	16384	394 [382, 406]	394 to 394	0/200	2.4e-01	48388 (40/40)

tables.md has every family at every size, with Cmax and coverage medians, p, Cliff's δ and the verdict.

4. The m = 3 baseline is a trivial family of solutions

At m = 3 the null's Cmax is about 3B (394 at B = 128), not a few units as at m = 2. The cause is the solution family P − P + R = R: (x, x, r) and its permutations solve S_4 = 0 for every r ∈ X, which is 3B − 2 ordered tuples. They are solutions of the system as written, so the primary metric keeps them (frozen, PROTOCOL.md §4), and gate G5 confirms every base element carries them. With them removed, Cmax_nt is 15–18 at B = 16, 21–24 at B = 32, 24 at B = 64 and 30 at B = 128 for the null and every structured family, against 630 to 48,006 for the control. They stay below B^{m−1} by a wide margin, so the reading does not depend on the choice.

5. Reading under the frozen rules

rule (PROTOCOL.md §5)	outcome
gates G1–G5 on the study binary	pass (gates.txt)
positive control fires at m = 2 in every instance	200/200
positive control Cmax ≥ 0.5·B² at m = 3 in every instance	160/160 (Cmax ≥ 2.6·B²; it also exceeds B²)
positive control coverage below the null at p < α_c	p = $1.7 \cdot 10^{-25}$ at all nine sizes
A/A split of the null, p ≥ 10^{-3}	pass, minimum 0.12
a structured instance with Cmax ≥ B^{m−1}	none of 1,800
family vs null, 90 tests at α_c = $1.1 \cdot 10^{-4}$	0 rejected

Verdict wording from the protocol: every structured family is **within the null** on Cmax and on coverage at every tested (m, B); the control is **above** the null on Cmax and **below** it on coverage.

6. What this shows and what it does not

- **Shows.** At these sizes the literal §4.2 count does not distinguish a subgroup coset, an arithmetic progression, an interval, a cubic image or a Dickson image from a uniformly random base of the same size, on the largest count or on coverage, and it cannot be fired by any of them. It is fired only by a base with additive structure in the group, which also destroys coverage.
- **Detectable effect.** With 40 instances against 400 the tests reach α_c at |δ| ≈ 0.37 and detect |δ| ≈ 0.46 with 80% power (protocol §5). Smaller differences are not excluded; the largest observed |δ| is 0.27.
- **Not shown.** Toy primes only (p ≤ $4.2 \cdot 10^{6}$), m = 2, 3 only, the five parameter distributions listed, and the isogeny tower's V is not among them. The count says nothing about the cost of finding a decomposition or about the solving degree of any presentation; that question is LEAD-ICP-pkm-tower-algebraic-oracle's and LEAD-ICP-digit-presentation-solving-degree's.

7. Disclosures

- A dry run (seed 1, small sizes) preceded the protocol and led to the trivial-solution metric and gate G5. Its instances are not used.

- Route: AGENTS.md §9a sends x86 experiments to Iso Lab on EKS Spot. The container that ran this has no AWS access, so the run was local (two threads on a four-vCPU host shared with another job). The claims are exact integer counts and do not use a clock; `wall_seconds` is operational only (298 s at `m = 2`, 3,941 s at `m = 3`).
- `commit.txt` records the binary SHA-256 `3061da3e…` and the freeze commit. `runs/m2/rows.jsonl` is in the artifact store (manifest beside it, sha256 `19ccf5ad…`).
- No independent `review-run` record exists yet.

8. Sources

`RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §2.3, §4 item 2 and its 2026-10-31 correction; `research/lever2_closure_20261031/REPORT.md` (the identity: mean closure count `C(B+m−1, m)/N`); in crypto-autoresearcher `KN-FIND-007` (the same identity and the small-multiples base's coverage collapse), `EV-FBG-001` (exact yield over 288 cells), `IDEA-20260801-021` (scheme degree of a size-`B` description); Petit–Kosters–Messeng, PKC 2016 (the factor-base definition, read from the authors' PDF). The summation polynomials follow Semaev (2004), which was not re-read here; `S_3` and `S_4` as implemented are checked against the group law by gate G2.

9. Reproduce

`cargo build --release --example exact_count_membership`; `exact_count_membership --selftest`; then the two commands in `PROTOCOL.md` §3 with `--seed 20261101`. `plot_counts.py` (a layout script, document rendering only) rebuilds `counts.svg`, `counts.png` and `tables.md` from `runs/m*/summary.json`; `build_report.sh` rebuilds this PDF.